DeepShovel 2.0: A Scientific Reasoning Network for Evidence-Grounded Verification and Discovery

Haoxiang Zhang
1227538797@sjtu.edu.cn
Shanghai Jiao Tong University
Shanghai, China

Jiaxin Ding*
jiaxinding@sjtu.edu.cn
Shanghai Jiao Tong University
Shanghai, China

ABSTRACT

Scientific conclusions are easier to assess and reuse when their supporting evidence, assumptions, and reasoning remain explicit. We present DeepShovel 2.0, an operational system that reconstructs the principal arguments of scientific papers and links these arguments across selected papers in the same research domain. For each conclusion, the system records its premises, assumptions, and conditions of use. These records link to original passages, data, tables, and figures. For a queried claim, the system retrieves supporting and challenging arguments together with the premises needed to interpret them. The generated answer links its claims to these arguments and their original sources. Research agents retrieve these arguments, inspect original sources, and run numerical analyses to propose claims, challenge their premises, and refine investigation procedures. To illustrate this workflow, a completed Codex-guided study evaluates a published seawater pH predictor on two segments of the GP15 Pacific transect. The demo shows how users inspect a paper's argument, assess claims under their stated assumptions, and examine a completed agent-guided investigation.

CCS CONCEPTS

• **Information systems** → **Digital libraries and archives**; *Retrieval models and ranking*; • **Computing methodologies** → **Knowledge representation and reasoning**.

KEYWORDS

scientific reasoning networks, argument reconstruction, evidence provenance, claim verification, retrieval-augmented generation, research agents

1 INTRODUCTION

Scientific conclusions depend on observations, assumptions, and inferential steps. To reuse a conclusion, researchers need to understand its evidence and limits. We illustrate these needs with two tasks: checking the assumptions behind a fault-pressure estimate and testing whether a seawater-pH predictor transfers to new observations.

DeepShovel 2.0 represents a reusable scientific argument as a conclusion together with its premises, assumptions, reasoning steps, and source links. It reconstructs each selected paper's research question, main claims, and inferential dependencies, then connects these arguments across selected domain literature. The network records when each conclusion applies and how it supports, qualifies, or challenges other conclusions. It supports two complementary operations: tracing a conclusion backward to its justification, and using source-supported premises to derive new, testable claims.

To make arguments traceable across scholarly resources, each reasoning step links to specific documents, passages, figures, tables, and data. Users follow these links from a generated answer to its argument and original evidence. The running system is at https://deepshovel.iiot-shxt.xyz. Code, predefined evaluation cases, an English walkthrough, and the research reproducibility package are released at https://github.com/spiderLL2002/deepshovel.

Contribution and positioning. GeoDeepShovel builds scientific databases from literature [1]. DeepShovel 2.0 extends literature reuse to the arguments connecting data, assumptions, and conclusions. Related work addresses claim verification with evidence rationales [2], inter-document argument relations [3], and graph-based synthesis [4]. ResearchAgent [5] and DeepEvidence [6] extend literature use toward iterative idea generation and evidence exploration. DeepShovel connects these tasks in three ways. First, each inference node links the premises and assumptions used together in one reasoning step to its conclusion. Second, the system checks whether these premises support the target claim under the user's conditions. Third, agent-proposed claims re-enter the same evidence-search and comparison process. Figure 1 illustrates condition checking through the friction, cohesion, and stress assumptions behind a pressure estimate.

2 BUILDING A DOMAIN REASONING NETWORK

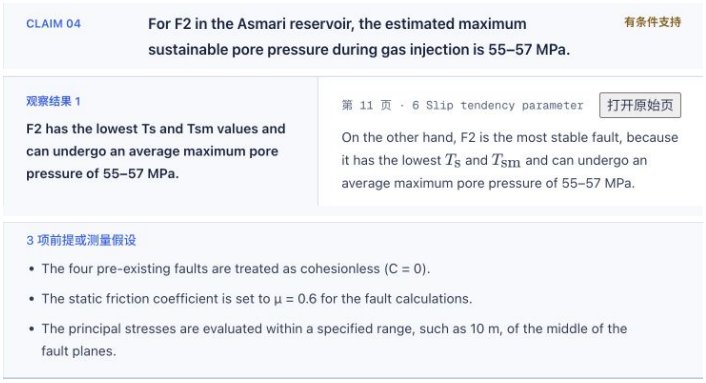
The system first parses papers and reconstructs their arguments. At query time, it retrieves candidate claims, expands their recorded dependencies, and compares the resulting arguments with the target claim. The browser exposes claims, their reasoning, and original evidence (Figure 1); an external agent orchestrates further source inspection and analysis (Section 4).

Running system. The deployed geoscience library contains 9,506 active papers, 41,092 claims, and 71,291 reasoning steps (29 October 2026). MinerU parses document layouts; current reconstruction and comparison use GPT-6-Luna. A curation interface supports paper selection and source-linked review. Queries compare up to 50 candidates and follow recorded dependencies to a maximum depth of 32.

*Corresponding author.

WWW '27, May 10–14, 2027, Dublin, Ireland
2027.

(a) Live inspector: claim, source, assumptions



Query friction	Recorded live relation
$\mu = 0.6$	equivalent
$\mu = 0.4$	qualifies

(b) Recorded dependencies behind this claim

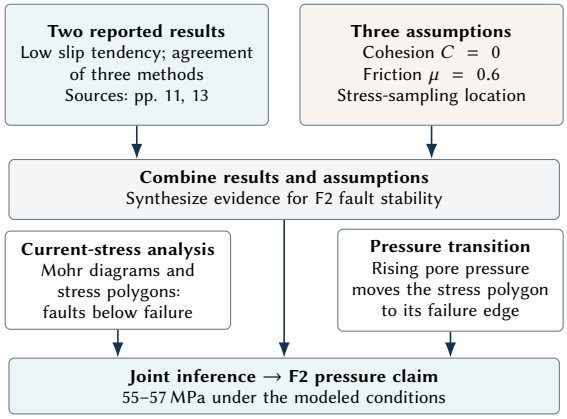


Figure 1: F2: (a) captured inspector states show the claim, linked source, and expanded assumptions; the table records live query relations, separately from the stored claim status. (b) The schematic separates reported results from the stress-state and pressure-transition reasoning. Arrows follow recorded inference inputs; the full expansion includes 18 premise links. An English walkthrough accompanies the demo.

Reconstruct the paper's reasoning. The parser extracts passages, equations, tables, figure panels, and supplementary data, recording their locations in the original document. Reconstruction follows the paper's main reasoning routes: question → methods and assumptions → observations → intermediate interpretations → conclusions. Users trace a conclusion backward to its observations and assumptions, or follow observations forward through intermediate interpretations to final claims.

To keep these connections explicit, we represent each reasoning step as

$$r = (P_r, A_r, w_r, c_r, \sigma_r, \pi_r). \tag{1}$$

Here, P_r lists observations and intermediate findings used together in the step. A_r lists its assumptions, and w_r describes in natural language how these inputs support conclusion c_r. The fields σ_r and π_r record applicability conditions and source locations. An assumption states what a derivation takes as given; an applicability condition records where its conclusion can be used. For example, a friction assumption also restricts the pressure claim to that modeled setting. Each inference node represents one reasoning step and groups the premises and assumptions used together to support its conclusion. Alternative derivations of the same conclusion occupy separate inference nodes. When a claim is opened, the system follows its stored links to retrieve the premises and assumptions used in each supporting step.

For fault F2 in the Asmari reservoir [9], the argument combines two reported analysis results—fault stability and agreement among three methods—with three assumptions (Figure 1). Current-stress analysis sets the initial state. Critical-pressure analysis then estimates when rising pore pressure would bring the fault to the modeled failure boundary. Users can inspect both the analysis and its original source.

Each reasoning step retains its applicability conditions and source locations. Conditions cover material, setting, method, comparator, and temporal or spatial range. Reconstruction distinguishes observations, interpretations, and predictions and retains quantifiers and negation. Source links identify a document version and a passage, equation, table, figure panel, or dataset record. Values retain units and uncertainty; figures retain captions and experimental context. Source passages remain accessible alongside reconstructed reasoning and agent proposals.

Connect arguments across papers. The same representation connects claims across papers while retaining their separate arguments and sources. For ecosystem-collapse timing at Fiskegrav, the system retrieves van Soelen et al.'s [10] 1.6–8 kyr estimate. It also returns Mettam et al.'s [11] evidence of continued animal disturbance of sediments after the last macrofossil. This supplementary argument distinguishes the disappearance of macrofossils from final ecosystem collapse. Users open both source pages to check which event the duration describes.

Model tasks and programmatic checks. Language models reconstruct reasoning steps and compare claims in context. Programmatic checks validate source identifiers and references; dependency traversal follows the stored links. This division connects source-grounded reconstruction with the query-time comparison described next.

3 REASONING-LEVEL RETRIEVAL AND VERIFICATION

Retrieve the justification of an answer. For each query, the system identifies the target claim, its applicability conditions, and the evidence needed to assess it. Lexical and semantic search retrieve candidate claims and observations. Dependency traversal then collects each candidate's premises, assumptions, and source evidence

into an *expanded argument*. A single model call uses these arguments to rank candidates and assign their relations to the target claim. Support and counterevidence searches cover incompatible observations, alternative mechanisms, failed transfers, and challenged assumptions.

During comparison, the search list displays a preview. The final relations distinguish equivalent conclusions from results that hold under different conditions. Users inspect the same expanded arguments and follow individual premises to their original pages.

After comparison, retrieval-augmented generation (RAG) produces an answer with explicit applicability conditions and links its assertions to reasoning steps and evidence.

Locate the disagreement inside the argument. Verification distinguishes equivalent claims, supporting evidence, contradictions, qualifications, and related evidence. Equivalence requires the same proposition under compatible conditions; a qualification limits where the result applies. A counterexample must address the claimed scope; evidence from another setting may instead challenge a generalization. The system identifies whether the disagreement concerns an observation, an assumption, or an inference. Dependency traversal connects a paper's principal claims, while alignment across papers exposes differences in data, methods, or applicability.

Evaluation. The retrieval evaluation contains 114 geoscience questions and 3,515 pooled judgments over lexical, dense, and hybrid results. The questions and datasets were reviewed by experts. For 104/114 questions (91.2%), at least one top-three hybrid result was judged to contain a complete answer (BM25: 105/114, 92.1%); this rises to 108/114 (94.7%) when partial answers count. BM25 reaches 93.0% under the latter criterion; nDCG@10 is 0.831 for hybrid and 0.895 for BM25. Hybrid covers two more queries when complete or partial answers count; BM25 has higher complete-answer coverage and nDCG@10. Retrieved evidence must also be assessed under the query conditions, which motivates the following comparison evaluation.

This comparison evaluation uses 30 fixed cases from five papers: matching claims, opposing conclusions, changed conditions, measurement attribution, and unsupported transfers. Before testing, GPT-6-Astra checked source passages and recorded acceptable relations, allowing multiple justified labels. We fixed the query, candidate, prompt, and GPT-6-Luna settings, varying the supplied evidence. Each run reuses the same 30 cases. The table counts predictions matching a predefined acceptable relation from candidate to query:

Comparison input	Run 1	Run 2	Run 3
Claim and conditions	28/30	28/30	29/30
Expanded argument	29/30	30/30	30/30
Source passages	29/30	29/30	30/30

In two public retrieval-and-comparison runs, predictions matched predefined acceptable relations in 29/30 and 30/30 queries. Median full-query times were 36.1 and 33.2 seconds. Inputs, outputs, and executable instructions are released. A separate source audit found 28/29 nodes and 23/24 dependency links fully supported across these five arguments.

Interactive condition check. For F2, a user opens the 55–57 MPa pressure claim and inspects the zero-cohesion, friction, and stress-sampling assumptions (Figure 1). The public system retrieves all three recorded assumptions for comparison. In two repeated query pairs, the original friction coefficient $\mu = 0.6$ yields an equivalent relation; changing it to 0.4 yields a qualification. Here, qualification identifies the mismatch with the source assumption $\mu = 0.6$: the reported estimate does not establish the pressure limit at 0.4. Users can inspect the same dependencies and source page 11 [9] behind that judgment.

4 AGENT-GUIDED DISCOVERY AND WORKFLOW REFINEMENT

An agent starts with a retrieved argument and uses its premises to formulate a testable extension. DeepShovel supplies the source-linked arguments and comparisons, while Codex coordinates source inspection, numerical analysis, and a record of candidate claims, their evidence, and their status. Codex accesses search and argument graphs through DeepShovel's HTTP API, then submits proposed claims as queries for counterevidence and prior work. We use task-level recursive self-improvement (RSI) for the agent's feedback-driven revision of its investigation protocol: the revised rule is saved and checked in the next analysis run.

An executed geoscience investigation. Across eight rounds, Codex screened 48 research questions for source support, novelty, and executable analyses. The selected study tested a published seawater-pH calibration [7] on two GP15 station groups. The calibration predicts pH from the slope of shale-normalized concentrations of four rare-earth elements: Sm, Gd, Dy, and Er.

DeepShovel supplied this rare-earth-element (REE) calibration argument and source; Codex defined a fixed-model test with public observations from the GEOTRACES GP15 Pacific transect [8]. REE and pH measurements were matched by parent-bottle identifiers, retaining their filtration status. The REE–pH predictor was compared with a depth reference based on the original calibration notebook. Both were evaluated against total-scale pH at 25°C and surface pressure, using the same eligible samples in six 100-m layers from 200 to 800 m. The depth reference uses the global pH mean for each layer saved in the original calibration notebook; neither predictor was refitted. The reported RMSE compares layer-mean predictions with layer-mean observations across the six layers. This comparison favored REE on Leg 1 and the depth reference on Leg 2 (Table 1). Within 200–800 m, the REE predictor therefore had higher RMSE on Leg 2. Figure 2 shows the data-matching and evaluation choices supporting this comparison.

Table 1: GP15 RMSE (pH units) across six equal-weight layer means. Leg 1: Seattle–Hilo; Leg 2: Hilo–Papeete. Both methods use identical eligible samples per layer.

GP15 group	Bottles / stations	REE model	Depth reference
Leg 1	24 / 6	0.070	0.170
Leg 2	55 / 8	0.218	0.110

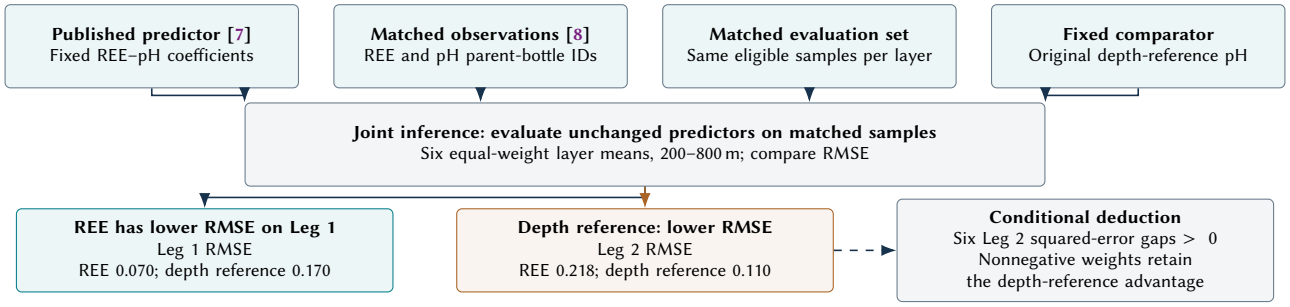


Figure 2: GP15 arguments: four joint premises support the comparison. The dashed arrow adds a check of all six layerwise errors to derive the reweighting claim for unchanged layer means.

The agent then derived a new conditional claim from the layerwise errors. For each of the six Leg 2 layers, the squared error of the REE predictor exceeded that of the depth reference. Let $d_i = e^2_{\mathrm{REE},i} - e^2_{\mathrm{ref},i}$. All six differences are positive, with a minimum of $0.00545\,\mathrm{pH}^2$. Therefore, $\sum_i w_i d_i > 0$ for any shared nonnegative weights summing to one: no shared nonnegative weighting of these six fixed layer means can reverse the depth reference's lower RMSE.

Codex records the comparison and deduction as distinct claims with data and analysis links. Codex submits each claim with its conditions as a new query for supporting evidence or competing explanations.

The GP15 study also illustrates task-level RSI. Agent review checked a revised rule for reporting station-bootstrap intervals: at least six original stations and all six depth layers in at least 90% of resamples. Codex saved and reran the protocol; 97.5% and 97.8% of resamples retained all six layers for Legs 1 and 2.

5 INTERACTIVE DEMONSTRATION

The demo follows three questions: What supports this conclusion? Does it apply here? What can we infer next?

1. Understand a published conclusion. Attendees enter the F2 pressure claim and open its source paper. They expand its observations, reasoning steps, and three assumptions, then open the original passage on page 11 (Figure 1).

2. Examine support and disagreement. Next, attendees change a condition in their target claim, such as the assumed friction coefficient for F2, or request counterevidence. The comparison assesses whether the recorded premises support this revised claim and presents the relation with its evidence. They follow each result through its premises to examine its applicability.

3. Extend the investigation with an agent. Finally, attendees follow the completed GP15 study from its research question to the comparison and conditional deduction in Figure 2. Using the configured Codex session, attendees can then propose another station group or request literature that could challenge the inference. During the live demo, a new request runs the initial literature search and planning steps. The saved GP15 study provides immediate access to the later analyses and recorded claims.

Conference setup and public materials. A laptop runs the browser and configured Codex session. Preview results remain visible during the roughly half-minute comparison. Internet access supports fresh retrieval and agent requests; the published study and saved source pages remain available for guided inspection offline. The repository provides the application, English walkthrough, and executable research package. In the reported rerun, all 76 reference files from the ten-stage GP15 analysis were reproduced byte for byte.

6 ETHICAL USE OF DATA

Publications and data are used under their licenses, with attribution, source versions, and provenance retained. The geoscience study reanalyzes public measurements; generated hypotheses remain candidates for further investigation.

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